

Probability Distribution

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Outlines



- **Recall** – Mean, Variance, Standard Deviation
- Histogram, Polygon, KDE
- **Recall** – Normal Distribution and Skewness
- Normal Dist, Standard Normal Dist, Z-Score
- (Additional) Data Imputation

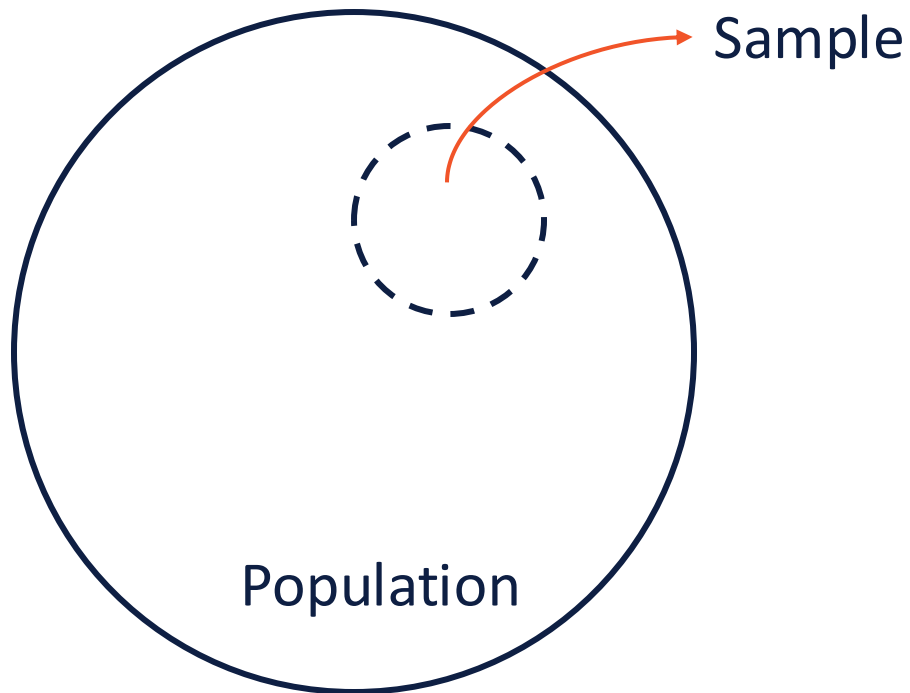


Recall – Mean, Median, Std. Deviation

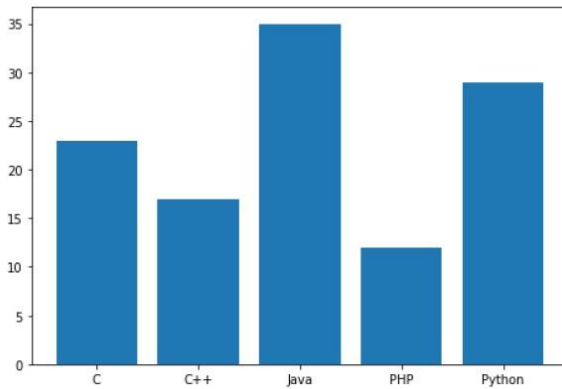
Recall the old concept that we already learned

Mean, Variance, Std. Deviation:

Population vs. Sample

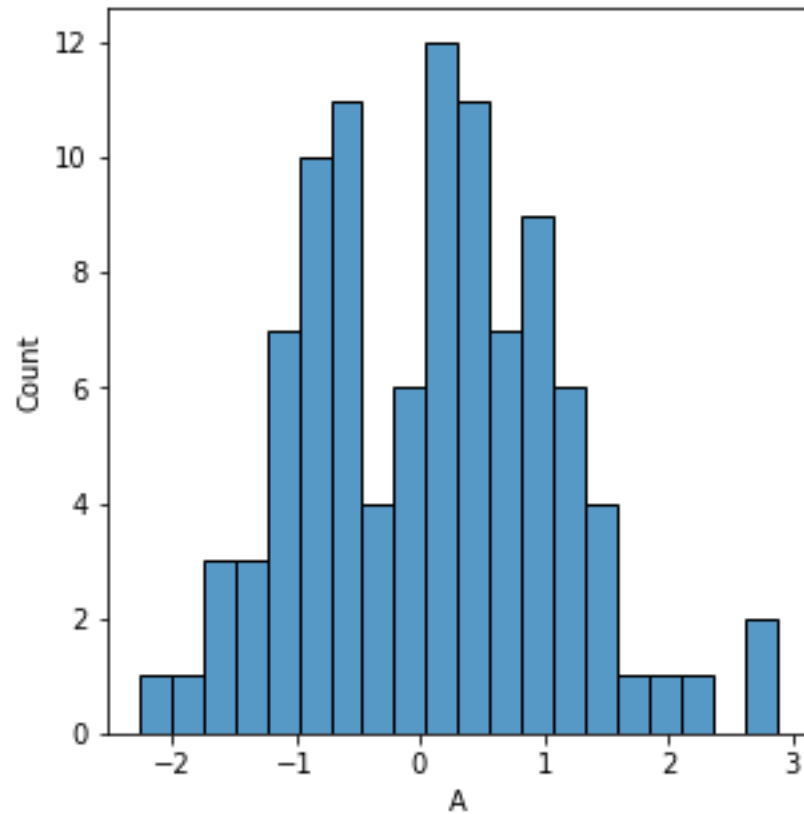


	Population	Sample
Data	N	n
Mean	$\mu = \frac{\sum x}{N}$	$\bar{x} = \frac{\sum x}{n}$
Variance	$\sigma^2 = \frac{\sum (\mu - x)^2}{N}$	$S^2 = \frac{\sum (\bar{x} - x)^2}{n - 1}$
Standard Deviation	$\sigma = \sqrt{\sigma^2}$	$S = \sqrt{S^2}$

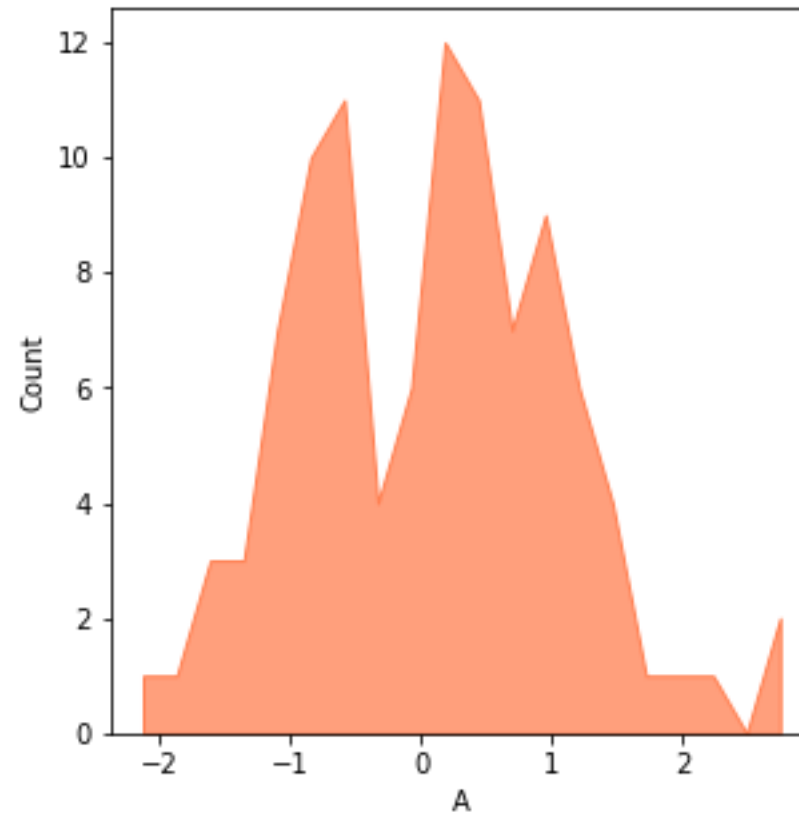


Histogram, Polygon, *Kernel Density Estimation (KDE)*

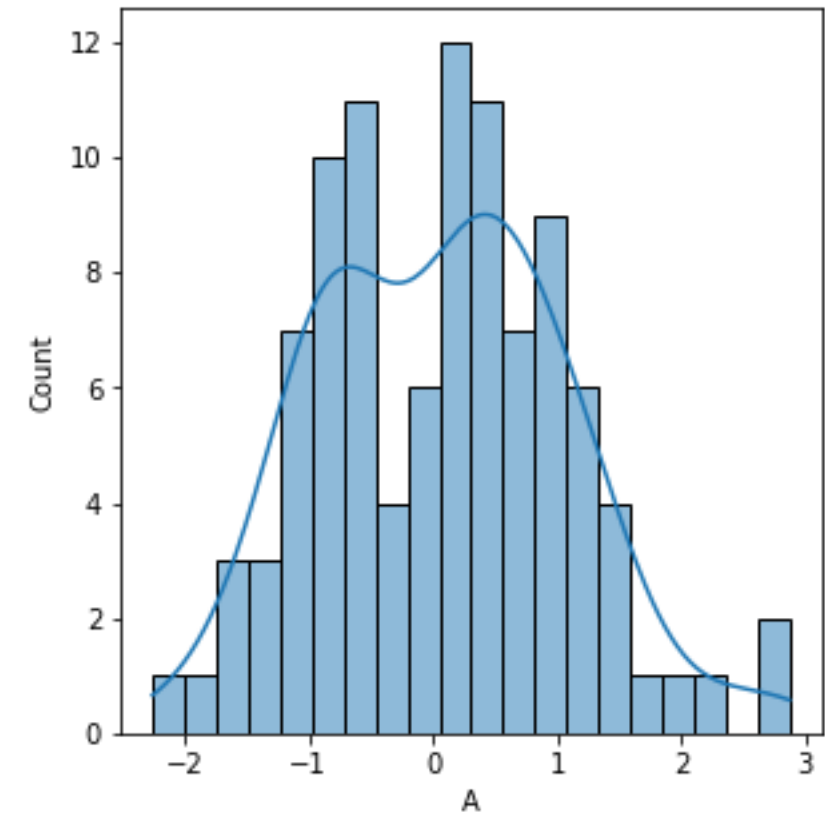
Who are they?



HISTOGRAM



POLYGON



**KERNEL DENSITY ESTIMATION
(KDE)**

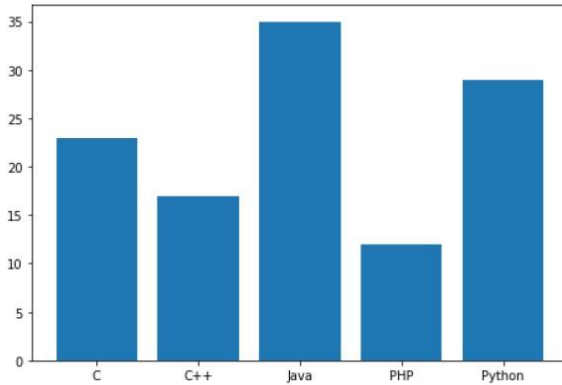
Histogram vs. KDE



	Histogram	KDE
Type	Bar chart with bins	Smooth curve
Data type	Discrete and continue	Continue
Parameter Sensitivity	Bin count/width	Bandwidth (smoothing)
Works well with	Small & large datasets	Medium & large dataset
Interpretation	Easy	Abstract
Shows Exact Counts?	Yes	Estimation
Best for	Frequency counts, comparisons	Smooth probability distribution

Histogram vs. KDE – Best Practice

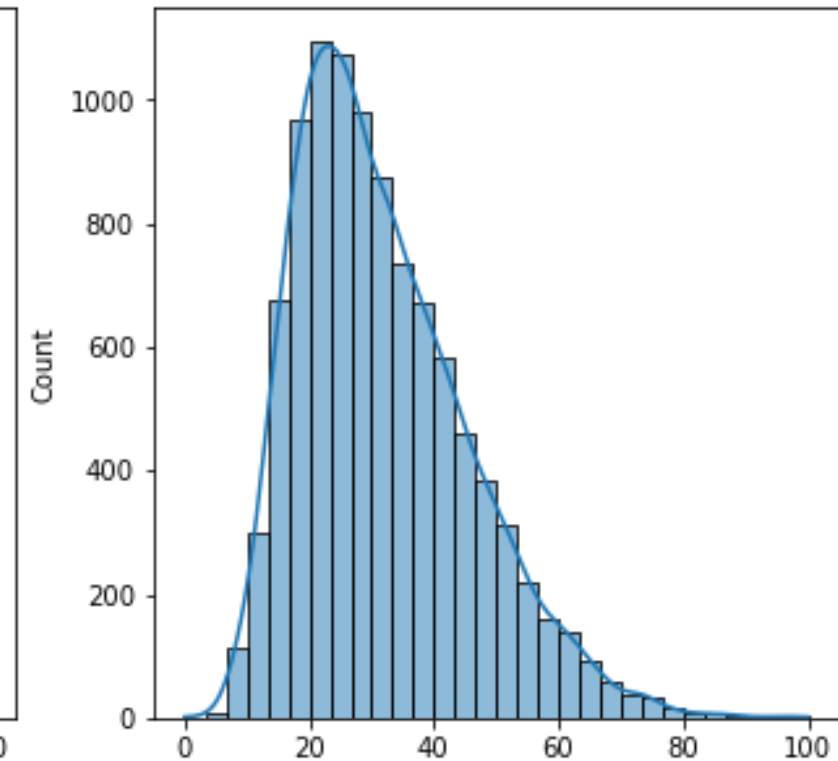
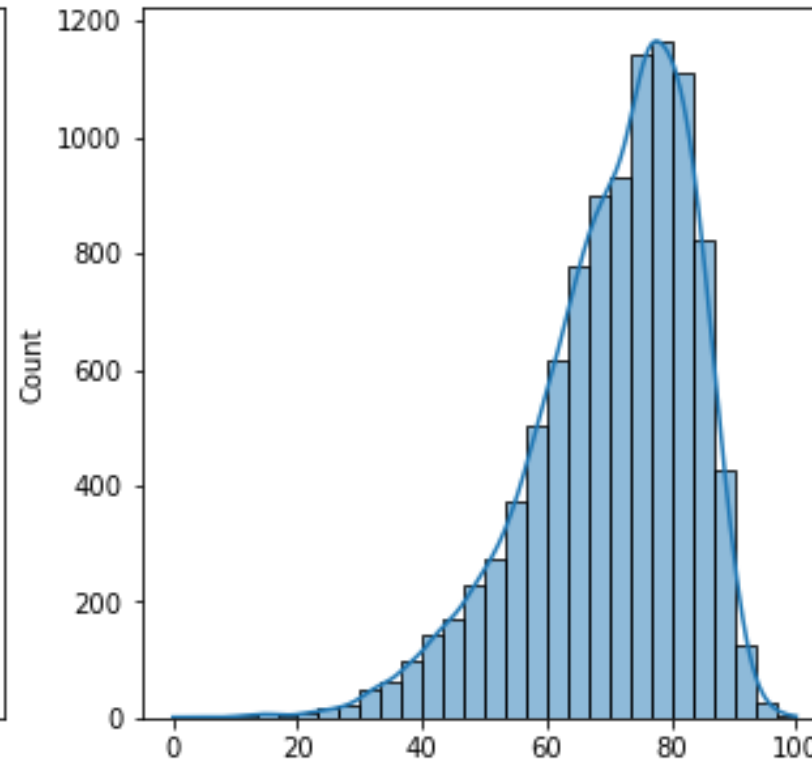
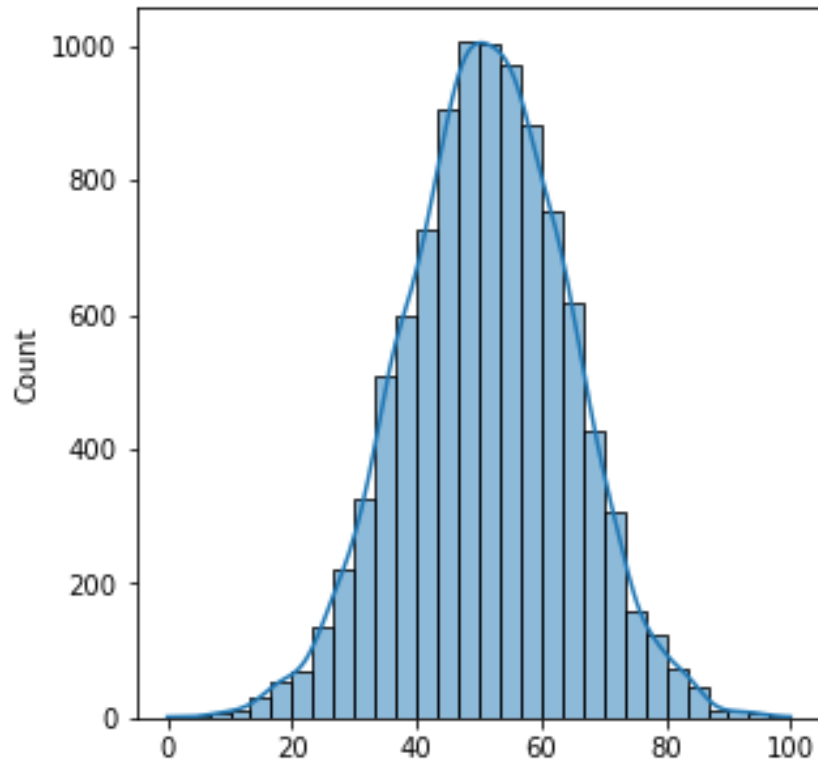
- **Survey Data (e.g., Age Groups of Customers)**
→ Use a **histogram** since the data is discrete and grouped into categories.
- **Income Distribution of a City**
→ Use **KDE** to see a smooth representation without arbitrary binning.
- **Comparing Test Scores from Two Classes**
→ Use a **histogram** if you want exact counts; use **KDE** if you want to compare distribution shape.
- **Small Sample of Heights ($n=10-20$)**
→ Use a **histogram** because KDE may not produce a reliable estimate.
- **Large Dataset of Product Prices ($n=1000+$)**
→ Use **KDE** for a clearer view of the distribution.



Recall – Normal Distribution and Skewness

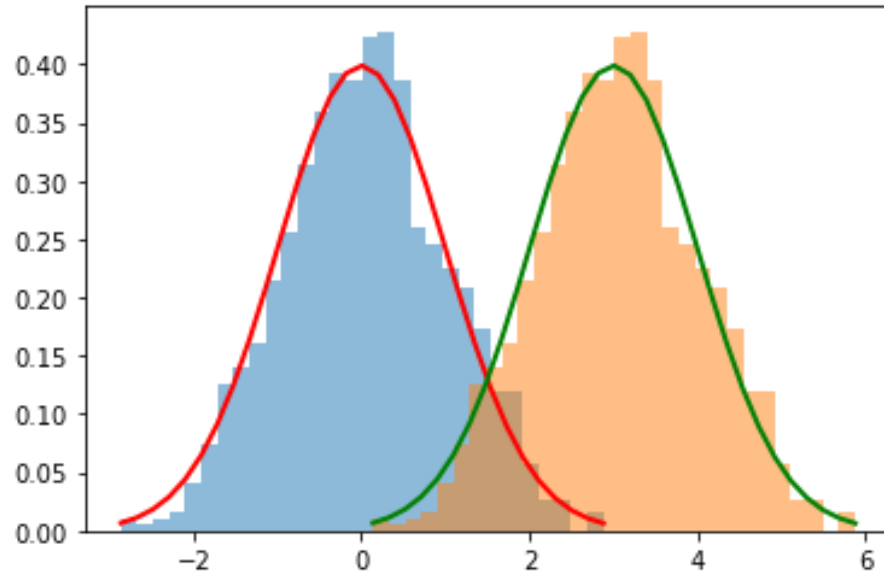
Check again, the relationship between data distribution and central tendency

Normal Distribution and Skewness



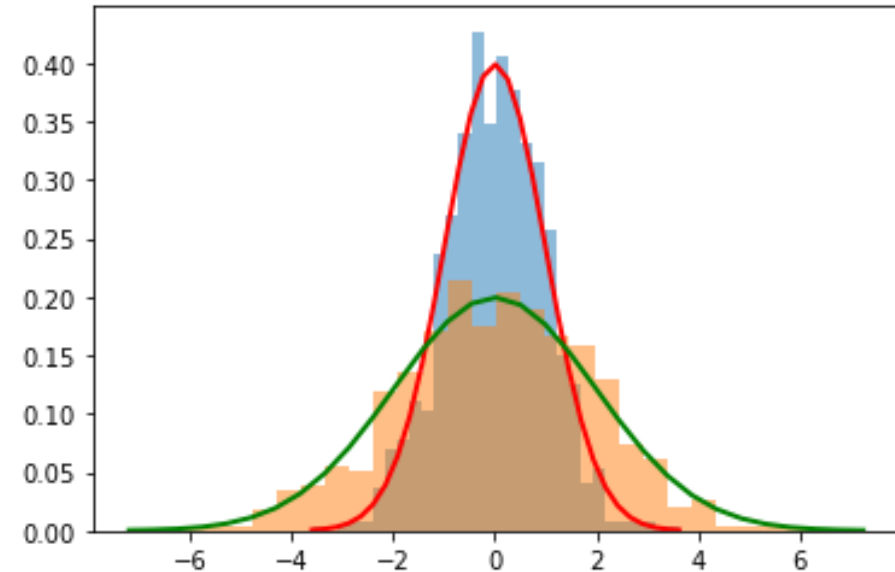
Do you still remember that data distribution patterns?
How data shifting and scaling affect to our data distribution?

Recall – *Shifting* dan *Scaling*



shifting

Mean = $x+k$
 Median = $x+k$
 Mode = $x+k$
 Range = x
 IQR = x
 STD = x



Scaling

Mean = xk
 Median = xk
 Mode = xk
 Range = xk
 IQR = xk
 STD = xk



Probability Distribution, Normal Distribution, and Z-Score

The pattern that tell you more about data

What is probability distribution?

- A **probability distribution** describes how the values of a random variable are distributed. It tells us the likelihood of different outcomes in a dataset.
- Among various probability distributions, **Normal Distribution** (also called **Gaussian Distribution**) is the most commonly used in statistics.

Normal Distribution

A **normal distribution** is a symmetric, bell-shaped probability distribution where most values cluster around the **mean (μ)**, and the probabilities decrease as you move away from the mean.

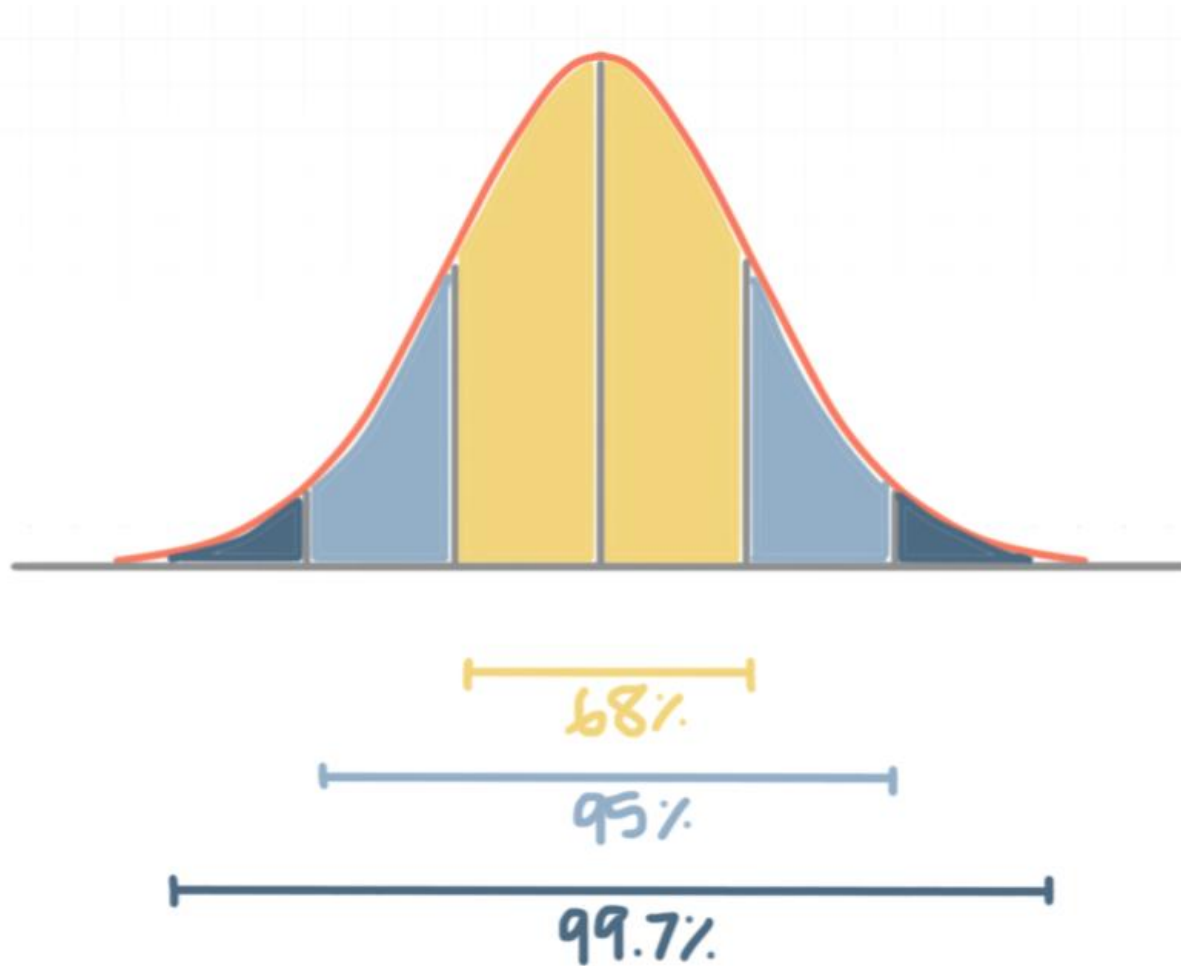
- The formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where,

- $e = 2.71828 \dots$ (rounded)
- $\pi = 3.14159 \dots$ (rounded)

The Empirical Rule (1)



What is *empirical rule*?

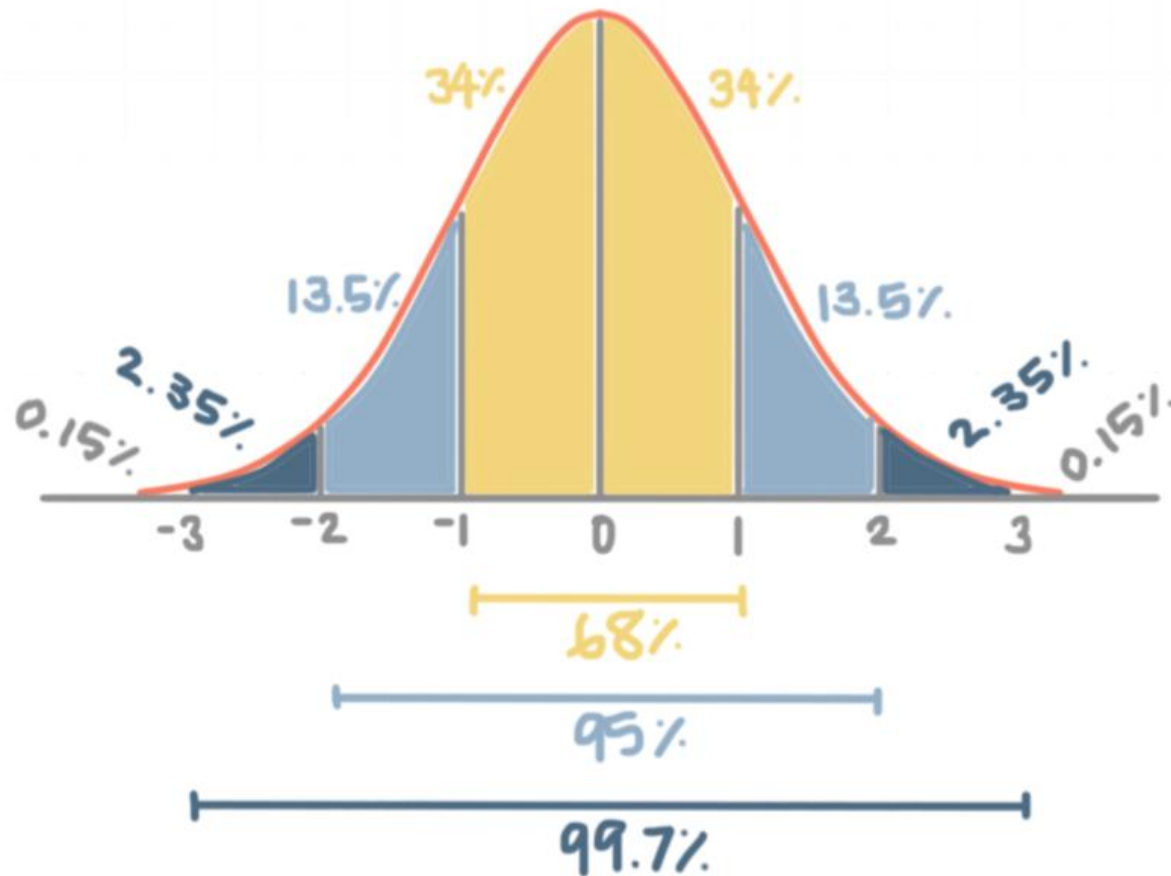
Every normal distribution follow the *empirical rule*, which are

- 68% data covered around 1σ
- 95% data covered around 2σ
- 99.7% data covered around 3σ

What about another 0.3%?

The Empirical Rule (2)

Therefore,



By applying this concept, we can determine the area under the curve for a given dataset using **percentiles** and **z-scores**.

Percentile

The **n-th percentile value** represents the value below which **n percent** of the area under the curve falls. Thus, if a value is at the **95th percentile**, it means that the value is greater than **95% of the other data points**.

Z-Score

The standard deviation (std) indicates how far the values deviate from the mean.

Percentile

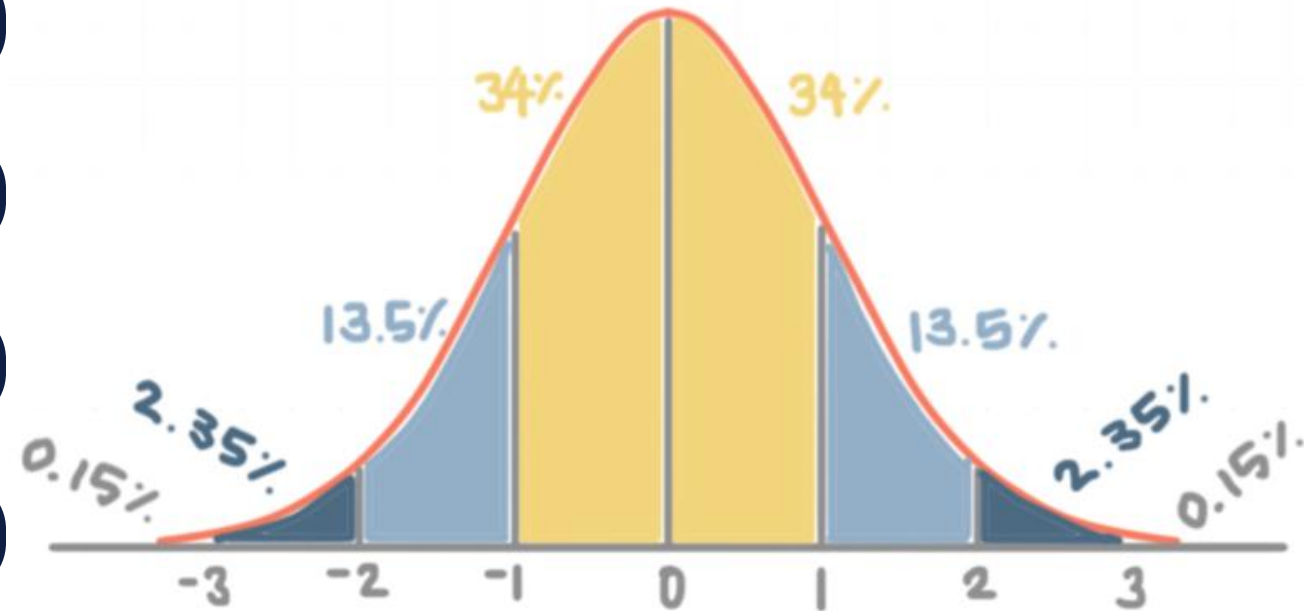
The Fact about *Percentile*

50th Percentile = Median = Q2

25th Percentile = Q1

75th Percentile = Q3

IQR = 75th Percentile – 25th Percentile

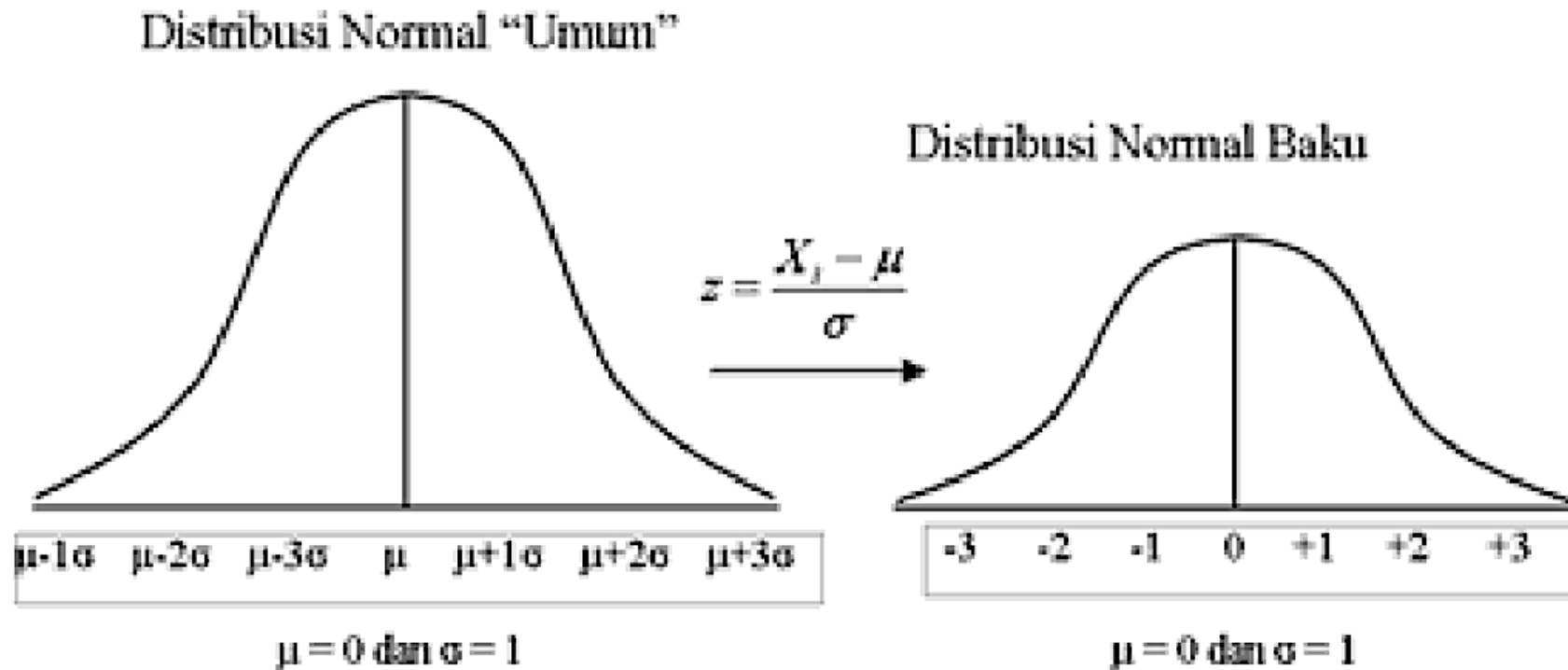


Standard Distribution

- The **normal distribution formula** and the **standard normal distribution (Z-score transformation)** are closely related. The **standard normal distribution** is simply a **special case** of the normal distribution where:
 - *mean, $\mu = 0$*
 - *standard deviation, $\sigma = 1$*
- Now, instead of using x , μ and σ , we rewrite the normal distribution formula in terms of Z ,

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}}$$

Normal Dist. Vs. Standardized Normal Dist.



Sumber: Statistika Terapan, Mikha Agus Widiyanto (2013)

Z-Score

The Formula Z-Score

$$Z = \frac{x - \mu}{\sigma}$$

z is z-score

x is the data

μ is mean

σ is standard deviation

Example

Give $x = 17.5$, $\mu = 16$, and $\sigma = 1$.

Compute the value of z ?

$$z = \frac{17.5 - 16}{1} = \frac{1.5}{1} = 1.5$$

So, it can be concluded that, 17.5 is pada 1.5σ away from the mean

What about the percentile of 17.5?

Z-Table: Relationship Between Percentile and Z-Score

Problem

Given $x = 17.5$, $\mu = 16$, $\sigma = 1$, and $Z = 1.5$. Which percentile x is?

Solution

USE Z-TABLE

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

So, the 17.5 is on 93.32 percentile

Case study

The average sprint time of athletes in the competition is **70.42 seconds**, with a standard deviation of **0.34 seconds**. How fast must an athlete complete the sprint to be faster than **95%** of the competitors?

Therefore,

The athlete aims to be in the **top 5% (0.05) of the best times**, meaning the fastest **5%** (with a shorter sprint duration) or within the **lowest 5% range below the mean**.

Solution,

Check the z-score on the negative value

z	.00	.01	.02	.03	.04	.05	.06
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594

$$z = \frac{x - \mu}{\sigma}$$

$$-1.65 = \frac{x - 70.42}{0.34}$$

$$-1.65(0.34) = x - 70.42$$

$$x = 70.42 - 1.65(0.34)$$

$$x = 69.859$$

Thus, the athlete must achieve a time of **69.859 seconds** to be in the **top 5%**.

Data Imputation

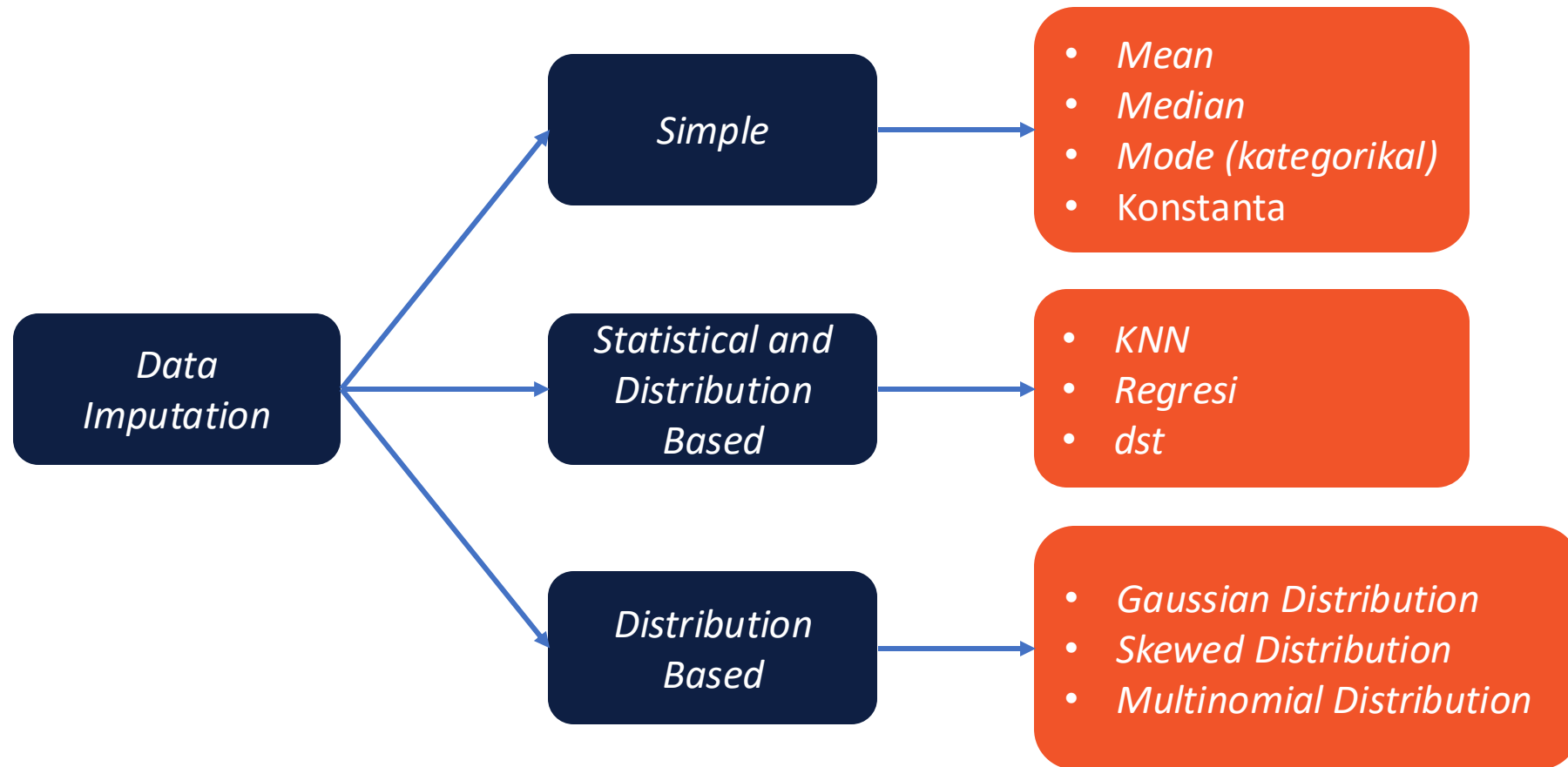
How to substitute the missing value in the datasets

What is *data imputation*?

Data imputation is the process of **replacing missing values** in a dataset with substituted values based on a **specific methodology**.

- Handling missing data properly is crucial because missing values can distort statistical analysis and machine learning models if left unaddressed.

Methods in *data imputation*



Case Study: Student Scores



Student	Math Score
A	85
B	90
C	NaN
D	78
E	Nan
F	88



Student	Math Score	Mean	Median	Mode
A	85	85	85	85
B	90	90	90	90
C	NaN	85.25	86.5	85
D	78	78	78	78
E	NaN	85.25	86.5	85
F	88	88	88	88

Assignment – Part 1

- Download dataset Titanic from <https://www.kaggle.com/datasets/yasserh/titanic-dataset/code>
- Check, whether the “Age” and “Fare” contain missing value or not. If yes, how many missing value on that variables?
- Check the distribution of “Age” and “Fare”.
- Use the correct simple data imputation method to impute the missing value on that variables (if any).

Assignment – Part 2

- Impute the missing value on “Age” using mean method!
- Compute the z-score of “Age” for all data.
- Compute the number of outlier from “Age” $\rightarrow |Z| > 3$
- Assume the “Age” is in normal distribution, What is the probabilities of “Age” < 20 ?

